BUS43-3

Valuation & Investment Analysis

DCF, multiples, scenarios, and the investment thesis

Pricing the firm as judgment under hypothesis sensitivity, not a spreadsheet recipe.

Albert School — 22 hours

Contents

Preface

You arrive at this course already able to do almost everything valuation requires — you simply have not yet pointed those skills at a price. From **BUS13-3** you can read the three financial statements as constructed documents and reconcile net income to cash. From **BUS23-3** you can decompose profitability with the SIG and DuPont, measure operational efficiency with ROIC, and — crucially — compute the free cash flow (FCFF and FCFE) that valuation actually discounts. From **BUS33-3** you can place a value on time itself: present value, NPV, the CAPM cost of equity, the after-tax cost of debt, and the WACC that blends them.

Valuation is the moment those threads converge on a single question: *what is this company worth, and should I buy it?* The mechanics — discount a cash flow, apply a multiple — you have largely seen. What is new, and what this course is really about, is **judgment**. A discounted cash flow model is a machine for turning assumptions into a number; the number is only as honest as the assumptions, and the assumptions are never certain. So we will treat every valuation as a hypothesis under test: state what must be true for the value to hold, isolate the two or three assumptions that move it, build bull / base / bear cases around them, and triangulate independent methods against one another. The deliverable of a valuation is never the number — it is the *argument*.

A single fictional company, **Meridian Components** — a mature industrial supplier — runs through the whole book. We value it by DCF, then by multiples, reconcile the two, stress its assumptions, and finally write a buy / hold / sell thesis on it. Watch the same firm look expensive through one lens and cheap through another: learning to say *why* is the entire skill.

1 Discounted cash flow valuation

Begin with the idea, not the formula. Imagine you could buy a vending machine that will spit out €100 of cash at the end of each of the next three years and then break. How much would you pay today? Not €300 — money next year is worth less than money now, because you could invest money now. If you require an 8% return, the machine is worth $\frac{100}{1.08} + \frac{100}{1.08^2} + \frac{100}{1.08^3} \approx 257.7$. That is a complete DCF valuation. A company is just a vending machine with an uncertain, growing, and (we hope) very long stream of cash.

DCF values a firm or an investment as the present value of the free cash flows it generates, discounted at a rate that reflects their risk. It applies the time-value machinery of BUS33-3 and the free-cash-flow construction of BUS23-3 to *pricing* rather than to a single accept/reject decision.

1.1 Which cash flow, and which rate

You learned two free cash flows in BUS23-3. The valuation question forces you to choose between them — and to pair each with its own discount rate.

> **Worked example — Meridian's free cash flow to the firm.** Meridian reports EBIT of €400m, a tax rate of 25%, depreciation and amortisation of €100m, capital expenditure of €130m, and an increase in working-capital requirement of €20m. Then
>
> $$\text{FCFF} = 400(1-0.25) + 100 - 130 - 20 = 300 + 100 - 150 = 250 \text{ m}.$$
>
> This €250m is the cash thrown off by the *business itself*, before any payment to lenders or shareholders.

Definition 1 (FCFF and FCFE) *Free Cash Flow to the Firm* is the cash available to all capital providers, before financing flows:

$$\text{FCFF} = \text{EBIT}(1-\tau) + \text{D\&A} - \text{capex} - \Delta\ \text{WCR}\ .$$

Free Cash Flow to Equity is what remains for shareholders after debt service and new borrowing:

$$\text{FCFE} = \text{FCFF} - \text{interest}(1-\tau) + \text{net borrowing}\ .$$

The two are discounted at different rates and yield different things:

Cash flow	Discount at	Gives	Then
FCFF	WACC	Enterprise value	subtract net debt → equity value
FCFE	cost of equity r_e	Equity value	use directly

Common pitfall The single most common DCF error is a **numerator–denominator mismatch**: discounting FCFF (a pre-financing, all-capital cash flow) at the cost of equity, or discounting FCFE at the WACC. FCFF is the firm's cash, so it is discounted at the firm's blended cost of capital and yields the value of the whole firm. FCFE is already the shareholders' cash, so it is discounted at the shareholders' required return and yields equity value directly. Pick one consistent track and stay on it.

1.2 Reinvestment discipline: growth is not free

Here is the assumption beginners most often smuggle in: they grow revenue at 10% a year while holding capex flat. That implies a company expanding output without buying machines or financing the extra receivables and inventory — a perpetual motion machine.

Definition 2 (Reinvestment and the cash growth costs) Growth must be paid for. The reinvestment that funds it — net capex (above depreciation) plus the increase in working capital — must be modelled explicitly and tied to the growth it produces. A useful discipline links the two through the reinvestment rate:

$$g = \text{reinvestment rate} \times \text{ROIC}, \quad \text{reinvestment rate} = \frac{(\text{capex} - \text{D\&A}) + \Delta\ \text{WCR}}{\text{EBIT}(1-\tau)}.$$

If Meridian wants to grow FCFF at 6% and earns a 12% ROIC, the identity says it must plough back 6%/12% = 50% of its NOPAT. Assume faster growth at the same reinvestment and you are implicitly assuming a higher ROIC — a claim that should be stated and defended, not hidden in a cell.

1.3 Putting it together: the explicit forecast

Definition 3 (DCF value) Over an explicit horizon of n years followed by a terminal value TV_n,

$$\text{EV} = \sum_{t=1}^{n} \frac{\text{FCFF}_t}{(1+\text{WACC})^t} + \frac{\text{TV}_n}{(1+\text{WACC})^n},$$

and analogously for equity value with FCFE discounted at r_e.

Worked example — Meridian's five-year forecast. Project FCFF growing at 6% from its €250m base, discount at a WACC of 8% (derived in Chapter 2), and use a terminal growth rate of 2.5%.

Year	1	2	3	4	5	TV
FCFF (€m)	265.0	280.9	297.8	315.6	334.5	—
Discount factor	0.926	0.857	0.794	0.735	0.681	0.681
PV (€m)	245.4	240.8	236.4	232.0	227.7	4243

The explicit five years contribute about €1{,}182m of present value. The terminal value (computed next) contributes €4{,}243m, for an enterprise value of roughly **€5{,}426m**.

1.4 Terminal value and its dominance

After the explicit horizon the firm does not stop; the terminal value captures everything beyond year n. Two standard methods:

Definition 4 (Terminal-value methods)

- *Perpetuity growth (Gordon)*: assuming FCFF grows at a constant $g < \text{WACC}$ forever,

$$\text{TV}_n = \frac{\text{FCFF}_n(1+g)}{\text{WACC}-g}.$$

- *Exit multiple*: apply a terminal multiple (e.g. EV/EBITDA observed today for mature peers) to the final-year metric, $\text{TV}_n = \text{multiple} \times \text{EBITDA}_n$.

For Meridian, $\mathrm{TV}_5 = 334.5 \times 1.025/(0.08 - 0.025) \approx 6\{,\}235$m, whose present value is $6\{,\}235/1.469 \approx$ €$4\{,\}243$m. That single number is **78%** of the entire enterprise value (Figure 1). The five years you laboured over are barely a fifth of the answer.

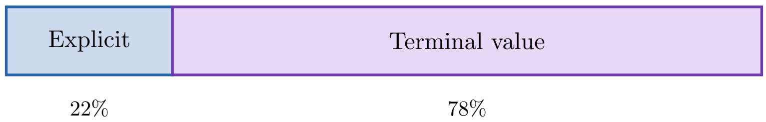

Figure 1: Composition of Meridian's enterprise value. The terminal value, which rests on the single hardest-to-defend assumption, dominates the output.

Common pitfall Because terminal value typically dominates, the valuation is mostly a bet on g and the discount rate, not on next year's sales. Two consequences follow. First, the perpetuity growth rate must **never exceed the long-run nominal growth of the economy** (a firm growing faster than GDP forever eventually becomes the economy) — a g of 5% is almost always an error, not optimism. Second, any terminal assumption must be stated, justified, and sensitivity-tested, because a 1-point change in g here moves the value far more than a 1-point change in any single forecast year.

1.5 From enterprise value to a price per share

FCFF discounting gives enterprise value — the value of the operating business. Shareholders own what is left after lenders are paid, so you must bridge across (Figure 2).

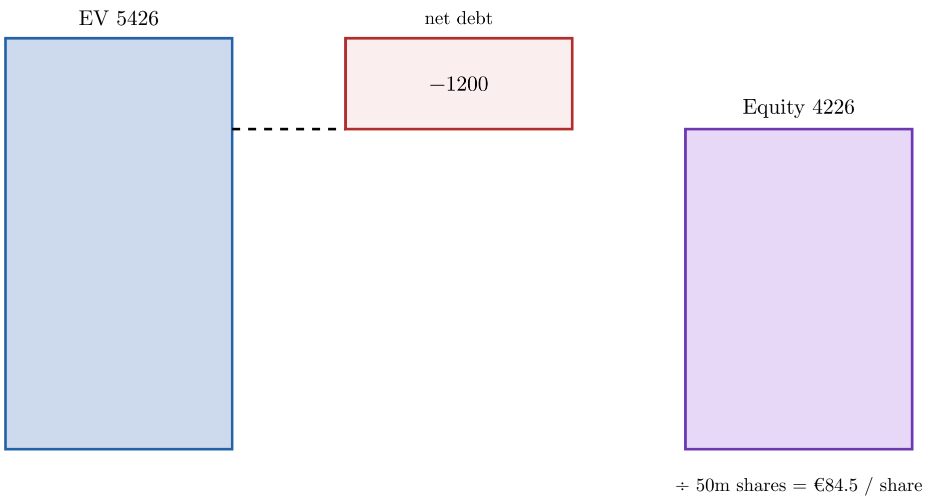

Figure 2: The EV-to-equity bridge for Meridian: subtract net debt (and any minority interests or other claims) from enterprise value, then divide by the share count to reach an intrinsic value per share.

With €1{,}200m of net debt and 50m shares outstanding, Meridian's equity is worth about €4{,} 226m, or **€84.5 per share**.

2 Cost of capital revisited

In BUS33-3 the WACC was a number you computed. Here it becomes a number you *defend* — and, more honestly, a *range* you defend. The machinery is unchanged; what changes is your relationship to its precision.

2.1 Re-deriving Meridian's WACC

Worked example — A defensible WACC. Cost of equity via CAPM: a risk-free rate $r_f = 3\%$, an equity risk premium of 5.5%, and a beta of 1.3 give

$$r_e = 3\% + 1.3 \times 5.5\% = 10.15\%.$$

Cost of debt: a pre-tax 5% with a 25% tax rate gives an after-tax $5\%(1 - 0.25) = 3.75\%$. With a target structure of 70% equity and 30% debt,

$$\text{WACC} = 0.7 \times 10.15\% + 0.3 \times 3.75\% = 7.1\% + 1.1\% \approx 8.2\%.$$

We round this to the **8%** used in Chapter 1 — and, as argued below, we should really carry it as a band of roughly 7.5–9%.

2.2 Why each input is contestable

CAPM critiques. CAPM assumes single-period, mean-variance investors pricing off one observable market portfolio. None of that holds cleanly, and realised returns are a noisy proxy for the required return the model wants. It remains the workhorse not because it is true but because the disciplined alternatives are not obviously better.

Beta estimation problems. A regression beta is the slope of the stock's returns against the market's (Figure 3). That slope drifts with the estimation window (2 vs 5 years), the return frequency (daily vs monthly), and the chosen index. It is unstable enough that practitioners often adjust it toward 1, or discard the regression entirely and build a *bottom-up beta*: take the unlevered betas of comparable firms, average them, and re-lever to the target's capital structure.

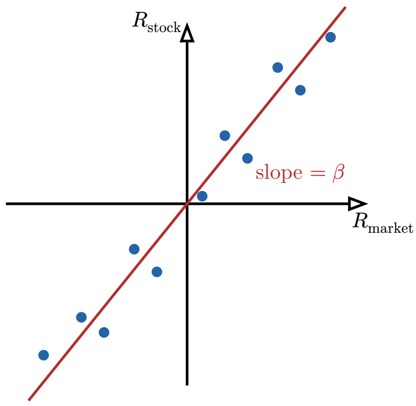

Figure 3: Beta as a regression slope of the stock's returns on the market's. The cloud is wide: the line you fit — and therefore the cost of equity — depends on the window, frequency, and index you happen to choose.

Country-risk adjustments. For cash flows exposed to riskier jurisdictions, add a country-risk premium to the cost of equity (or build the risk into the cash flows themselves), reflecting sovereign and currency risk a developed-market premium misses. Do it in one place only — adding it to both the rate and the cash flows double-counts.

Common pitfall The cost of capital is an **estimate, not a measurement**. Every input — the premium, the beta, the structure — carries a range, and the ranges compound. A WACC quoted as "8.23%" claims a precision the inputs cannot support; the honest deliverable is a band ("≈ 7.5–9%") and a statement of which input drives the width. Crucially, recall from

Chapter 1 that this rate sits in the denominator of the terminal value, where it is most leveraged — so false precision here is most expensive precisely where it is least warranted.

3 Trading and market multiples

DCF asks "what are the cash flows worth?" Multiples ask a different, faster question: "what does the market pay for companies like this one?" The two are independent lenses, and a serious valuation uses both.

3.1 The idea, then the catalogue

A multiple scales price by a value driver, so that companies of different sizes can be compared on the same ruler. If mature suppliers like Meridian trade at 7.5 times EBITDA, and Meridian earns €500m of EBITDA, the market would price its enterprise at $7.5 \times 500 = €3\{,\}750$m — without any forecast at all.

Definition 5 (The standard multiples)

- *EV/EBITDA* and *EV/EBIT* relate *enterprise value* to operating profit, and so are independent of capital structure.
- *P/E* relates *equity price* to net earnings.
- *P/B* relates equity price to the book value of equity.

Common pitfall Match the numerator to the denominator's **claimholder**. Enterprise-value multiples sit over pre-financing metrics (EBITDA, EBIT) because enterprise value belongs to all capital providers. Equity-price multiples sit over post-financing metrics (net income, book equity) because price belongs to shareholders. EV/net-income or P/EBITDA mixes the two and is meaningless.

3.2 The comparable set is the valuation

A multiple is borrowed credibility: you are asserting your target deserves what its peers get. That assertion is only as good as the peer set. Comparables should share the target's business model, growth, margins, and risk — not merely its industry label. A "media" bucket that lumps a streaming platform with a regional newspaper produces an average that describes neither.

Worked example — Three lenses on Meridian disagree. Apply the peer medians to Meridian:

- *EV/EBITDA* at 7.5×: $\text{EV} = 3\{,\}750 \rightarrow$ equity $3\{,\}750 - 1\{,\}200 = 2\{,\}550 \rightarrow$ **€51 / share**.
- *P/E* at 14× on EPS of €5.10: ≈ **€71 / share**.
- *DCF* (Chapter 1): ≈ **€84 / share**.

Three defensible methods, three different answers. Resolving this spread — not picking one — is the work of Chapter 8.

Common pitfall Multiples mislead whenever the comparison is not like-for-like:

- **Cyclicality**: a peak-earnings denominator makes a stock look cheap and a trough-earnings denominator makes it look dear, even at a constant price.

- **Growth differentials**: a faster grower deserves a higher multiple; comparing across growth rates without adjustment misprices systematically.
- **Comparable-set impurity**: peers with different leverage, accounting policy, or segment mix contaminate the benchmark — the very reason BUS23-3 taught you to adjust financials before comparing.

4 Capital markets structure and pricing

A valuation produces an estimate of intrinsic value; the market produces a price. Where and how that price forms shapes how far the two can drift, and what premiums and discounts you must layer on.

Definition 6 (Where securities trade) The *primary market* is where securities are first issued and the firm actually raises capital; the *secondary market* is where already-issued securities change hands between investors; *OTC* markets trade bilaterally, dealer-to-client, rather than on a central exchange.

- **IPO mechanics.** An initial public offering carries a firm from private to public ownership. A price range is set, demand is gathered through book-building, and an offer price is struck — typically below the first day's trading price, the well-documented "IPO underpricing" that compensates investors for taking unproven paper.
- **Efficient Market Hypothesis.** The EMH holds that prices already reflect available information, so consistent mispricing is hard to exploit. Treat it as approximately true: it is the reason your "buy" requires an explicit argument for *why the market is wrong*, yet limits to arbitrage, illiquidity, and behavioural effects leave real room for price to diverge from value.
- **Liquidity discount.** A stake that cannot be sold quickly without moving the price is worth less; private or thinly traded equity carries a discount of commonly 20–30%.
- **Control premium.** A buyer acquiring control pays above the trading price for the right to direct the firm's cash flows and policy — the mirror image of the minority discount a small stake suffers.

Remark These adjustments connect valuation to the price you can actually transact at. The €84.5 DCF value is for a marketable, minority share; valuing a controlling block, or an unlisted family firm, would add a control premium or subtract a liquidity discount respectively.

5 Non-standard valuations

The default DCF assumes a firm with positive, forecastable cash flows, a stable structure, and a clean separation between operations and financing. Push any of those assumptions and the recipe breaks. Each case below is treated at introductory depth — enough to recognise the trap and reach for the right adjustment, not to master it.

Business type	What breaks, and the adjustment
High-growth	Near-term FCFF is negative and meaningless; nearly all value sits in the terminal year. Use a long, explicit horizon until growth and margins

Business type	What breaks, and the adjustment
	normalise, and discipline the terminal state with a credible path to steady-state economics.
Cyclical	Earnings at a single point misstate normal earnings. *Normalise* over the cycle (Figure 4) — use mid-cycle margins for both DCF and multiples — rather than annualising a peak or trough.
Distressed	Going-concern value must be weighed against liquidation value and a real probability of default; WACC and capital structure are themselves unstable, so a single discounted path is fragile.
Financial institution	Debt is raw material, not financing, so FCFF and EV multiples do not apply. Value the equity cash flows directly; lean on P/B and on return-on-equity-versus-cost-of-equity, with regulatory capital as a binding constraint.
Platform / network	Value rests on user growth, engagement, and unit economics that precede accounting profit. Supplement — do not replace — the cash-flow logic with metrics that track the network and its monetisation.

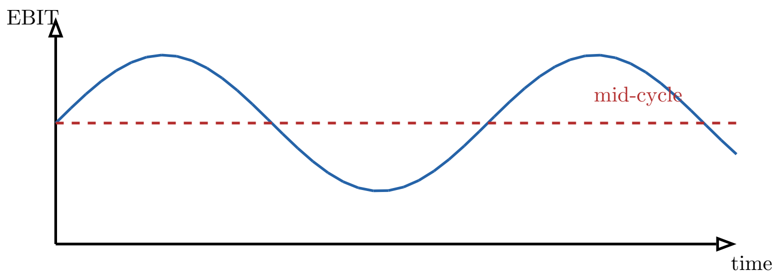

Figure 4: Normalising a cyclical firm. Valuing off the peak (or the trough) of the swinging earnings line embeds a temporary state forever; the dashed mid-cycle level is the defensible base for both DCF and multiples.

6 Scenario, sensitivity, and stress-testing

A valuation is a function of its assumptions. Reporting a single number hides that fact; reporting the function honours it. This is the discipline that turns a spreadsheet into judgment, and it must run *through* the valuation, not sit in an appendix.

Definition 7 (Bull / base / bear) Every valuation deliverable carries three internally consistent cases: a *base* case of most-likely assumptions, and a *bull* and a *bear* case built from **coherent** upside and downside *stories* — not an arbitrary ±10% nudge of every cell at once.

The key word is *coherent*: in a real recession Meridian's volume, margin, and reinvestment move *together*, so a bear case shocks them jointly rather than one at a time.

Worked example — What moves Meridian's value. Flex the two assumptions that matter and hold the rest:

	Bear	Base	Bull
Revenue growth	3%	6%	8%

	Bear	Base	Bull
WACC	9%	8%	7.5%
Value / share	≈ €58	≈ €84	≈ €105

The €58–€105 spread *is* the valuation. A point estimate of €84 that hides it is less honest, not more precise.

Definition 8 (Sensitivity analysis) Sensitivity analysis isolates the two or three assumptions that dominate the output by varying each in turn and recording the effect on value. For most going concerns these are the revenue growth rate, the operating margin, the terminal growth rate, and the discount rate.

Common pitfall Two failure modes. **Spurious symmetry**: shifting every input by the same ±10% treats independent and dominant assumptions alike and disguises which one actually matters. **Incoherent scenarios**: a "bull" case with recession volumes but boom margins describes no possible world. A good scenario tells a story a sceptic would recognise.

7 Real-options intuition

A static DCF discounts a *single* expected path. But managers are not passive: they can wait, quit, or double down as uncertainty resolves. That flexibility has value a point estimate cannot see.

Worked example — The option to wait. Meridian can build a plant for a new product now, with an expected NPV of zero. A naïve DCF says "indifferent." But demand will be revealed in a year: if it is high the plant earns +€200m, if low it loses €150m, fifty-fifty. By *waiting*, Meridian builds only in the good state and walks away in the bad — capturing the +€200m upside while flooring the downside at zero. The right to wait is plainly worth something, yet the zero-NPV "build now" calculation prices it at nothing.

Definition 9 (Real options) An *option to delay* defers an irreversible commitment until more is known; an *option to abandon* exits and recovers residual value if outcomes disappoint; an *option to expand* scales up if outcomes are favourable.

The common thread is **convexity** (Figure 5): when you can keep the upside and cut off the downside, more uncertainty raises value rather than lowering it — the opposite of the intuition a single discounted path trains.

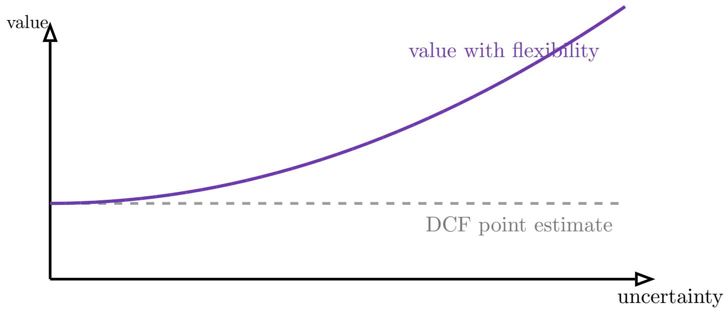

Figure 5: Why a DCF point estimate understates an irreversible-but-flexible decision. The flat line is the single-path DCF; the convex curve is the value once managers can delay, abandon, or expand. The gap is the option value.

Common pitfall Real-options intuition is a reason a *low or zero* DCF can still justify action — it is **not** a licence to inflate every valuation. It applies only where the decision is genuinely irreversible *and* the future genuinely uncertain *and* management genuinely has the flexibility to respond. Absent those, "but there's option value" is hand-waving.

8 Triangulation across DCF and multiples

You now hold three numbers for Meridian — €84 (DCF), €51 (EV/EBITDA), €71 (P/E) — and the temptation is to average them into €69 and move on. Resist it. The average destroys the only information that matters: *why they disagree.*

Definition 10 (Football-field analysis) A *football-field* chart displays, for one target, the value range implied by each method — DCF across its scenarios, each multiple across its peer range — as horizontal bars, so the overlaps and the divergences are visible at a glance (Figure 6).

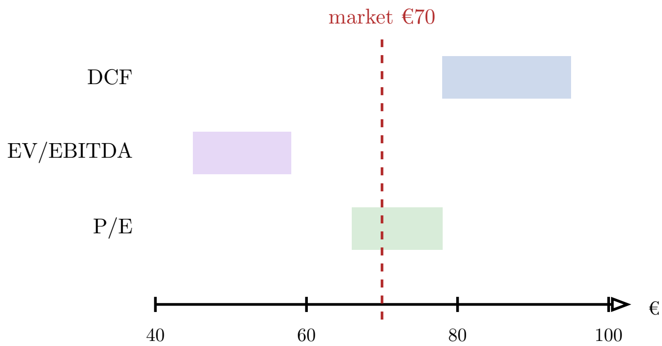

Figure 6: Meridian's football field. DCF sits well above the EV/EBITDA range and the current price; P/E straddles the market. The valuation's output is the explanation of this pattern, not a midpoint.

The reconciliation argument. Read the field as a set of claims. The EV/EBITDA range is low because peers are mature and barely growing; the DCF is high because it bakes in 6% growth and a 12% ROIC; the P/E sits near the market price because trailing earnings already reflect today's reality. So the entire case reduces to one question: *is 6% growth real?* If it is, the market (and the EBITDA multiple) are anchored to a stale past and the stock is cheap; if it is not, the DCF is a castle in the air and €70 is about right.

Common pitfall The output of a valuation is the reconciliation **argument**, not the range and certainly not the average. "The methods give €51 to €95, midpoint €73" is a non-answer. "They diverge because DCF prices growth the multiples ignore, and that growth hinges on the new-product ramp" is a valuation. Averaging independent estimates that disagree for a knowable reason throws away exactly the reason you were paid to find.

9 Investment thesis construction

Everything converges here. The course terminates in a complete investment thesis on a real, named public company, defended in Q&A against an invited practitioner juror. The thesis is where mechanics become a recommendation.

Definition 11 (The investment memo) A professional investment memo states a **buy / hold / sell** recommendation and defends it with a fixed spine:

1. *The business* — what it does and the economics of how it makes money.
2. *The valuation* — DCF and multiples triangulated, with bull / base / bear scenarios.
3. *The thesis* — the two or three assumptions the recommendation depends on, stated as falsifiable claims.
4. *The risks* — what would make you wrong, and what you would watch.
5. *The catalyst* — why the gap between price and value should close, and roughly when.

Worked example — Meridian, assembled. Triangulated intrinsic value ≈ €84 in the base case (€58–€105 across scenarios) against a market price of €70 — about 20% upside. The recommendation is **Buy**, but the memo is honest that the entire edge rests on one claim: that Meridian can sustain 6% growth via its new product line, which the market, pricing it like a flat mature supplier, currently doubts. The bear case (3% growth) lands at €58, *below* today's price — so the thesis is a bet on the growth assumption, sized to the confidence it can honestly carry.

The recommendation follows from the gap, between triangulated intrinsic value and market price, sized against the confidence the assumptions support. A narrow, well-supported edge can still be a "hold"; a wide edge resting on one heroic assumption may warrant only a small position.

Common pitfall A thesis is judged not by the elegance of its point estimate but by how well its author knows which inputs drive the conclusion and how far it survives their plausible range. The fatal answer in the jury room is "the model says €84." The strong answer is "it's €84 *if* 6% growth holds; here is why I believe it, here is what would change my mind, and here is what the stock is worth if I'm wrong." Valuation, finally, is judgment under hypothesis sensitivity — and the defence is where that judgment is examined.

The arc of this course. Build the cash flows (BUS23-3) and discount them at a defended cost of capital (BUS33-3) to a DCF value; cross-check against what the market pays for honest comparables; recognise where the standard recipe breaks; wrap every number in coherent scenarios; respect the flexibility a point estimate misses; triangulate the lenses and argue the divergence; and deliver a buy / hold / sell thesis whose strength is its author's grip on the two or three assumptions that decide it.